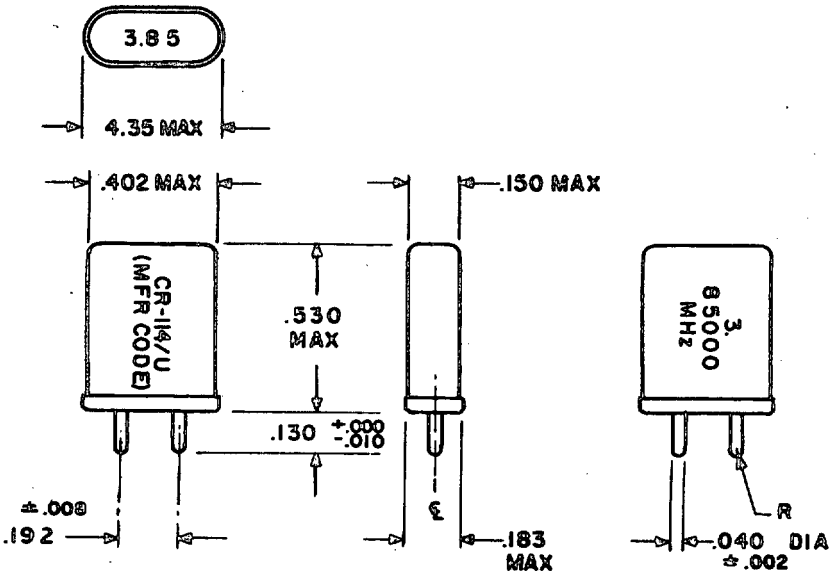
MILITARY SPECIFICATION SHEET
CRYSTAL UNIT, QUARTZ, CR-114/U

The complete requirements for procuring the crystal units, quartz, described herein shall consist of this document and the issue in effect of MIL-C-3098.



INCHES	MM	INCHES	MM
.002	.05	.183	4.65
.008	.20	.192	4.88
.010	.25	.402	10.21
.040	1.02	.435	11.05
.130	3.30	.530	13.46
.150	3.81		

NOTES:
1. All dimensions in inches
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm to the nearest .01 mm
3. The type-designation, frequency, and code-designating characters shall not be less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width

FIGURE 1 Outline dimensions and marking example for crystal unit CR-114/U

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TABLE I. Detail Requirements.

Requirement	Specified Value or Condition
Holder	HC-25/U, modified (MIL-H-10056/11)
Dimensions and marking	(See Figure 1)
Frequency range	2.9 to 3.85 MHz, inclusive
Frequency tolerance:	
Operating temperature range	±0.005 percent
Mode of oscillation	Fundamental
Operating temperature range (noncontrolled)	-55° to +105°C, inclusive
Temperature run - operating	Method A
Government reference standard test set	TS-330/TSM
Capacitance, load	32.0 ± 0.5 pf
Rated drive level	1.0 ± 0.2 mw
Calibration values	Resistance ..... 100 ohms Crystal current .... 3.2 ma
Capacitance, shunt	7 pf, maximum
Equivalent resistance	300 ohms, maximum
Shock: 1/	
Frequency change permitted	±0.0005 percent
Equivalent-resistance change permitted	±15 percent
Vibration: 2/	
Frequency change permitted	±0.0005 percent
Equivalent-resistance change permitted	±15 percent
Unwanted modes	Minimum unwanted mode effective resistance shall be at least twice that of the main mode or greater over the operating temperature range -55° to +105°C.
Aging:	
Frequency change permitted	0.0005 percent

1/ Method 202 of MIL-STD-202 (MIL-C-3098, Method A).

2/ Method 204 of MIL-STD-202, test condition A, 0.10 inch double amplitude or 5 gravity units, whichever is less.

3/ Requirements not applicable: Reduced drive level; Heat cycling; Bond strength.

4/ These crystal units are for use in the AN/ARC-51 equipment. They were previously fabricated using Air Force Technical Order T.O. 12-1-20 and/or Collins Radio Company's specification 289-5599 through 5618-00.

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Air Force - 85

Review activities:
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